

ZIXUAN FANG

zixuan.fang@monash.edu

EDUCATION

Monash University

PhD in Econometrics and Business Statistics

Melbourne, Australia

Expected Graduation: 2028

The University of Manchester

Master of Mathematical Finance

09/2022 - 09/2023

Distinction

Xi'an Jiaotong-Liverpool University (XJTLU)

Bachelor of Financial Mathematics

09/2018 - 07/2022

GPA: 3.78 / 4.00

TEACHING EXPERIENCE

Monash University

Teaching Assistant – Foundations of Quantitative Finance

07/2025 - 11/2025

- Facilitated tutorial sessions, effectively explaining complex quantitative finance concepts and mathematical models to students.
- Provided comprehensive academic support during office hours, resolving student inquiries regarding course materials and problem-solving techniques.
- Assessed student assignments and examinations, providing constructive feedback to enhance learning outcomes and ensure adherence to academic standards.

RESEARCH EXPERIENCE

Distinguishing Risk Attitude and Strength of Preference

Supervisor: Athanasios Pantelous

07/2024 - Present

- Investigated the theoretical distinction between risk attitude and strength of preference, addressing the limitations of traditional utility models that conflate these fundamental components.
- Utilized simulation analysis to quantify the impact of narrow framing on decision-making, demonstrating that neglecting strength of preference leads to significantly inflated estimates of risk aversion.
- Introduced the concept of “relative risk attitude” via the transformation $u(v(x))$ to rigorously disentangle pure risk aversion from value functions.
- Proposed a refined framework for preference assessment with practical implications for improving the accuracy of financial behavior modeling and portfolio allocation.

Practical Application of Momentum Strategy in the Futures Market

(Suzhou Financial Intelligence Technology Co., Ltd.)

Supervisor: Xuanyang Yin

06/2022 - 08/2022

Description: Code Showcase

- Conducted research on futures product strategies, specifically testing momentum strategies on the Chinese financial platform - Myquant.
- Implemented a robust approach by sorting monthly returns of all futures products.
- Developed opening conditions: selecting winning and losing portfolios, initiating long positions in the winning portfolios, and short positions in the losing portfolios.

- Implemented risk control measures by monitoring the portfolios in real-time. If a winning (losing) portfolio falls below (rises above) the 9-day moving average, promptly replaced them with top-ranked (bottom-ranked) futures products.

High-Frequency Data Processing (XJTLU)

Supervisor: Dr. Yi Hong

06/2021 - 06/2022

Description: Code Showcase

- Designed and constructed a high-frequency data warehouse in SQL for the derivative securities market in China.
- Conducted strategy studies on securities products (stocks, futures, and options) trading based on high-frequency data.
- Empirically confirmed the existence of a contrarian effect in Chinese futures markets during bull markets, and a momentum effect during the COVID-19 pandemic.
- Sharpened practical skills in using high-frequency data for capital allocation, strategy development, and risk management.
- Examined the profit of momentum strategies over the pandemic period, achieving a Sharpe ratio of 5.

INTERNSHIPS

Bohai Bank

Position: Assistant Manager

08/2021 - 09/2021

- Gained critical understanding of Chinese banks' operations, workflows, and organizational structure.
- Assisted in processing clients' information into the bank's internal system.
- Analyzed development trends within the banking sector in China.

Haitong Securities

Position: Assistant Manager

07/2019 - 08/2019

- Securities Market and Client Services: Excelled in brokerage and private capital services; utilized Python for multi-dimensional client analysis and tailored product recommendations.
- Business Processing & Client Services: Assisted 30+ clients with Sci-Tech Board access and resolved 100+ account issues, ensuring high client satisfaction.
- Financial Auditing & Document Management: Managed transaction audits and data categorization, identified irregularities, and mastered basic accounting documentation.

SKILLS

- **Programming & Tools:** Python, MySQL, JAVA, MATLAB, R, VBA
- **Software Proficiency:** Microsoft Office Suite (Excel, Word, PowerPoint), LaTeX
- **Languages:** English, Mandarin